

ActInf GuestStream #008.1 ~ "Exploring the Predictive Dynamics of Happiness"

GuestStream | 1:39:54

hello everyone and welcome to the active inference lab

today it is august 9th 2021 and we're here in guest stream number 8.1 with mark miller and guests so this is going to be an awesome presentation followed by a discussion so if you have any questions during the presentation please put them in the live chat and we'll make sure to ask the panel at the end of the presentation so thanks again to all of our great guests for coming on and please take it away thank you excellent thanks for that daniel okay

i'm just going to share my screen oh can you uh turn on screen sharing for me on your side

um it's under the advanced so at the bottom where it says share screen there should be a little little arrow and click that there should be an advanced feature and then should say allow got it

okay go for it yeah that's it uh-huh great how's that can you just see my slide does that look right

excellent great okay well daniel thanks thanks so much man for the invitation to come and give a talk i mean we've been trying to get we've been trying to find an opportunity to speak together for you know a couple of months now well since i moved to japan so i'm excited to be here and i'm excited to be talking about this topic which we're um we're really

excited about it's a new paper that
actually julian kiverstein who's here
right now and myself and eric reitfeld
um have a preprint out right now in the
the paper title title of predictiveness
of happiness and well-being we're also
anticipating that um inez hippolyto and
laura sinbad smith will be here this
evening lars has had a slight emergency
so um he's on the road but we're hoping
he'll be able to call in um because
actually the work that uh we feature in
this paper

it builds on or it's really closely
related with the research that anez and
lars are doing so the idea was that
we're gonna be able to get um we're
gonna be able to really talk to them
about their individual portions so
hopefully they're on the stream by the
time we hit that place and uh if it
turns out that they're not that's okay
we can just go through and then we can
chat at the end um if you don't know me
uh my name is mark miller

i'm an assistant professor at the
university of hokkaido um in their
center for human nature artificial
intelligence and neuroscience and i am
uh an active inference researcher
right so

let's see

um

right so i guess probably lots of people
who are listening to the stream tonight
and definitely the guests who are here
right now hey anez great you made it
right on

um know that um recently there's been a huge amount of research coming out that's linking um using the active inference framework in order to investigate all sorts of psychopathologies you know mostly finding a home in computational psychiatry circles but really we're we're seeing active inference accounts come up you know relating to depression and addiction and ocd and schizophrenia and dissociative disorders and all sorts and while there is this sort of active inference boom right now in relationship to computational psychiatry and in particular psychopathology we're not seeing any work really talking about um how we might apply active inference frameworks to the theoretical and formal approach of discussing things like well-being so you know all those topics that would naturally fit in something like a positive psychology are still primarily being overlooked although you know um in all fairness we are starting to see it come up at the edges because we're starting to see people talk about the benefits of hallucinogens and also the benefits of meditation so that was part of our motivation in writing this paper was to see if there was anything interesting to say in a positive account yeah not just the negative account but the positive account

um but the other reason why we decided to write this paper was because actually julian and eric and i have written a series of papers now on psychopathologies in particular using the active inference framework so we have papers out now on um depression and mood disorders and addiction and depersonalization and ocd and we started to see that there were some shared characteristics between these various psychopathologies if you look at them through the sort of computational lens of active inference and seeing those shared characteristics sort of started providing we think clues about you know what might be going right when a human is happy healthy and well and that's what our paper is about and that's what this guest stream will be about as well looking a little bit more at what some of the predictive dynamics might be that are underlying um subjective happiness and overall well-being and of course it's a big topic so we're not trying to exhaust you know and try to exhaust the whole of what well-being would be but rather just taking a sort of first step to see if we can work out some of the some of the points that might be interesting so since i have these lovely people here tonight with me um i've decided to make my presentation relatively short um i hope it won't be any more than sort of 30 minutes all said and um i'm going to pitch it at a

relatively high level of abstraction

because i think there's like a number of

nice moves i'd like to give to the

audience here tonight to sort of provoke

research and provoke some ideas and um

if you're interested more in the nitty

gritty um we have the paper out in

pre-print so you're more than welcome to

go and actually check out the paper and

of course if any questions come up we

can talk about that as well

great

okay so

so i think a sort of natural place to

start thinking about the relationship

between active inference and well-being

is to think about optimality um in

cognitive neuroscience and psychology

today well-being is often spoken of as a

form of optimal psychological

functioning um but what exactly optimal

means is not very well explained usually

or at least it's

underdeveloped and of course

as a free energy

principle and active inference framework

researcher when you look at that you the

first thing i thought was well there's a

natural way to sort of be thinking about

optimality um in terms of the active

inference framework right free energy

minimization over the long run

approximates optimal active inference

right that is sort of bayesian inference

is a form of optimal belief updating so

um

that fact in some way already gives us

um

you know it gives the free energy principle in the active inference framework you might think a sort of normativity already you know predictive systems tend to create or strive towards an optimal fit between their phenotypical self-organization and the sort of statistical regularities that they encounter in their environments so um we think that an important part of well-being from this perspective is going to have something to do with being a good predictor of the sort of hidden causes so that's both in the environment and inside the agent of the sort of sensory states that we tend to encounter and of course that crucially relies on our being able to optimally update our generative model in ways that are life-sustaining and flourishing inducing and i don't think that that's an incredible stretch just as an opening gamut um you know we know that well-being is closely related with health in various um various layers of health you know mental health and emotional health and physical health and good predictors they tend to self-organize in ways that allow them to maintain homeostasis across various contexts and over the long term right so um oh and you know and sort of best of all um if we think along these lines which i think could be fruitful lines to think along than we have at our

disposal as lots of um the listeners
today already know this sort of ever
evolving and ever-growing mathematical
suite to help us describe what is a good
model what's a good model all about so
that's our starting point our starting
point was to think well maybe well-being
could be discussed broadly
as a kind of optimal updating yeah and
the goodness the goodness of the
generative model that the agent uses to
form beliefs and control their actions
in life-sustaining ways over the long
term
okay so that's a good start
and i think it fits very nicely with
what we already know about lots of
psychopathologies in particular that
psychopathologies are characterized by
sub-optimal beliefs and the sorts of
behaviors that come from having
sub-optimal beliefs um depression is a
good example it's just one out of a hat
though really i think if you look at any
psychopathology you're going to find
some sort of bad belief is at play at
some level of the hierarchy i think
depression is a nice example you know
recent accounts of depression and not
just from the act of inference camp um
are highlighting the role that rigid
beliefs about one's inability to manage
the complexities of their life including
especially the social complexities of
their life is playing a major role in
depressive symptoms and we've written a
paper on this um pretty recently
and um in that paper we using the active

inference framework we argued that when an agent is met with enough persistent volatility then they can eventually develop a high level prior about their own high level prior belief about their own um inability to manage volatility and when you get uh that high level prior installed something pretty nasty can happen right um the system a predictive system then starts selectively sampling the world to try to confirm that hypothesis and in fact it down regulates any sort of counter evidence that goes against that deep prior that deep pathological prior so so that even when the situation changes you know if the agent finds themselves in more positively predictable situations uh they tend to hold on to that belief in volatility and that's really the what we think is the pathological moment so another way of saying that is an agent at least a depressed agent yeah has uh become overly confident that it is uncertain it's overly confident about being uncertain we're not the only people who've said that clerking colleagues have written a fantastic paper um definitely check it out if you don't know it and in that they come to the same conclusions they write in this sense we might conjecture that major depression occurs when the brain is certain that it will encounter an uncertain environment okay and um as we all know you know due to the central role that prediction

plays in sculpting our experience and directing our behaviors of course as soon as we have a high level bias like this installed um then it has tremendous power to

produce exactly the outcomes that would um that would confirm those biases it's one of the problems of being a predictive agent you know the predictive system is really good for lots of things but one of the problems is there's this sort of deep self-fulfilling prophecy that can occur okay

um

the rigidity of these bad beliefs is especially important here and it comes up again when we start thinking about what well-being might be um you know there's mounting evidence that cognitive rigidity you know the sort of stickiness of bad beliefs um but they're a basic feature of psychopathologies and in fact you know um some people are even saying that it might be the root cause of all psychopathologies and um i don't think we disagree um there's a nice aeon article about this if you uh if you're interested in that point and i think i might just uh take a take a little break there and hand it over to anez um and this hippolyto has a really just a fantastic forthcoming paper coming out that everyone should be excited to be reading which offers um a really beautiful computationally rich account of how insulated internal states yeah when these bad beliefs sort of get stuck and

then they resist updating how that
creates um the sorts of pathological
belief systems the pathological
behaviors that we see in psychopathology
so um if it's okay and as i'm just gonna
stop share and let you maybe talk a
little bit about insulated bad beliefs
and then i'll pick it up from here and
keep going on to well-being you know is
that okay

yes for sure um hi everyone can you hear
me and see me

yeah okay great so first of all thanks
so much for mark for inviting me to be
here and it's always such a pleasure to
be here with daniel uh for these
wonderful sessions and uh hi julian
uh very nice to see you as well
so i'll just now share my screen
if i think that's possible yeah
let me see

okay so for some reason i'm not able to
share my screen

i
think it should maybe try again

okay

no it doesn't go through for some reason
uh

strange it works for me fine

i have no idea

we could go for a uh

an audio

thought or you can try to share it

later let me just leave and come back if
that's okay maybe that's good you know
the trick

um this is uh this is this is the new
world of technology isn't it you have to

do everything on the fly you have to do
everything on the fly and be willing to
adapt

i i have one interim question

yeah please yeah i hit it

were you working on active inference or
interested in those approaches and then
came to be studying

psychopathology or just well-being more
broadly or was it the other direction
like which way did you come at this
intersection from

yeah you're right and um it started the
other way around so um in my masters in
which julian was my advisor at the
university of edinburgh in 2010

um the last chapter of my senior thesis
was a proposal so i was working on
empathy looking at um um you know
philosophy and cognitive science of
empathy looking at dynamic systems
theories of emotion the last chapter was
on predictive processing and active
inference and then i got to do my phd
with andy clark

specifically on active inference and
then it grew out of doing that into
psychopathology yeah we got it very good
we got it excellent we got it some
technical difficulties but uh we are now
good okay all right you can see it fine
yeah yep nice work thanks and let's
continue perfect

okay so um all right so um yes this work
is on uh uh what we call psychotic macro
blankets and this is a something that
i've been working on uh for the last few
months with two different groups

applying this to uh different um
uh two different things so one
specifically schizophrenia specifically
and the other one is um
disembodiment so what i thought i would
do here today would be to do like sort
of like a combination between um the two
so that we get like a bigger picture of
what uh we have in mind with them with
uh with applying this particular
framework

so um

yeah so um

let me just uh give some like framing
about uh where this comes from and uh
why and to motivate it a little bit so
that was the parts from this concept
that mark also just mentioned the
self-organization which is super
relevant to these things that we are
talking about when we refer to
well-being and mental health and all of
that um

and as a few words about that it's an
emergence of order from disorder and
specifically without an intervention
from the external intervention that is
external to the system

um and then what is important here for
us is to think of self-organization as a
pattern or an arrangement that gives
rise to a specific way in which the
system is self-producing and
self-distinguishing and here and not to
the inactivist literature

so but the idea here that the
fundamental concept is the
self-organizing pattern that that comes

up

um

this is why why are we talking about
that because we want to look at how
the systems or the organisms as a whole
are

interacting with the environment in a
way that has been called in embodied
literature as a tournament with the
environment right

so the idea there is that these
organisms or the systems are a tune with
their environment and uh in a way that
they will show a certain specific
pattern um or part of the arrangement
that will be distinct from for example
systems that are not attuned to the
environment and this is very well known
in

the whole bulk of literature in this
embodiment and phenomenology

so uh the idea that we want to pursue
here is actually relating to a little
bit too complex systems theory in which
we want to say that

this differentiation between embodiment
and disembodiment are different patterns
it doesn't mean that the system
that the system

that is not adjusted to the environment
should be considered as disordered but
instead that it has a different kind of
order

right

so he would be sort of like rejecting a
little bit the computationalist accounts
of psychopathology as simply a disorder
a disorder or an illness and

opening up a little bit of more like a
broad scope or spectrum of um to talk
about mental health
um so then if we set up things like that
then uh what needs to be explained is
how uh this self-organizing behavior of
situated organisms come to be ordered in
a certain way rather than another and
then we'll be uh now talking about uh
mental health

so um then we think that active
inference is um a very interesting and
useful

tool precisely to
think about uh this uh self-organization
and the patterns that come from the
self-organization
um we think about um active inference as
a model of

uh leaving things adjust to the
environment and we all know what active
inference is so i'm just gonna highlight
what is relevant here for us in terms of
pattern dynamics

we think that it is a nice and cool
model that
when associated with tools such as macro
blankets or renormalization group it can
be applied to many different scales of
self-organizing activity and that's
where we find all of these patterns of
our interest of our scientific interest
so it is useful to explain um these
gradients spectrums of kinds of patterns
of order in self-organizing systems so
just highlighting these ideas
um so we take it as a model um in this
case so it means that we are not

committing to the
to the realist view of
that let's say that the phenomenon
studied under active inference
ontologically possess or leverages the
conceptual tool but that the tool is
really important and useful for us to
understand uh certain phenomena in
this case particularly what you want to
understand is this embodiment
right so we all know active inference
but again let me just highlight uh just
some important points to understand than
the psychotic markov blankets later on
so um

here the idea is that self-organizing
self-producing self-distinguishing
system will correspond to the internal
states which affect and which effects
and affect and are affected by the
environment and the external states
via a set of other states such as the
active and the sensory states
active and sensory states are then
constructed as influencing one another
giving rise to a certain pattern of
self-organization and this is what's
going to be relevant for later on
so in a way what we want to say is that
well an adjusted system
that self-organizes is a system that um
minimizes the free energy which we all
know can also be understood as
uncertainty or entropy and so on right
but an adjusted system would be
something like that
a system that is just well adjusted to
its environment and for technicalities

obviously there are these references on the bottom

now moving on to this embodiment and then later on applying we want to apply this

active inference strategy to

further highlight or understand what is going on with this embodiment um

just a few words about the disembodiment literature

so the idea is that comes from

phenomenology and a little bit from

existentialist philosophy and the idea

is that the primary experience of the

body is the body experienced as a

subject and this is really crucial

um

the thing is that this experience of the

body as a subject can be lost and

in which case the body is inspected as

an object more like a side of scientific

interest and here is what uh the

literature says that that's when this

embodiment comes up is when the body all

of a sudden ceases to be experienced as

a subject and is now inspected as an

object

so what happens is uh in exploring and

interacting with the local environment

we usually typically we do not attend to

our body we just like take it for

granted got our body that allows us to

access move around interact explore the

world through our body

and it is not until we cannot do these

things or something is wrong that we

will direct our awareness towards the

body

and then the body becomes salient as an object of inquiry and it's right here at this point where we find this embodiment it's when we cannot um explore or interact with the world as one would like would like so in a way we can um distinguish embodiment and disembodiment by saying uh for example or just highlighting these features of course the literature is vast but i think that these are the ones that work for us here um embodiment is an attunement with the environment that is allowed or enabled by having an experience of the body as a subject um and that gets us attuned um to the environment through a bodily self-awareness or something that there is a bodily self-awareness that is that is uh felt then this embodiment on the other on the other hand is a disatnment with the environment because there is a loss of the body as a subject body becomes an object there's a disatnment and a diminished body awareness so this comes from the literature and it is interesting for us here why because then there's been these links that have been made between psychopathology and psychedelics and disembodiment as in this embodiment being something that is reported in different psychopathology such as schizophrenia the

personalization

even major depression

or other

psychopathologies that involve

rumination

and another very important thing is that

this embodiment also occurs or is what

is reported in the encyclopedias intake

and

this has been um even more relevant uh

studying this embodiment precisely in

the psychedelic renaissance that we are

witnessing today and also another point

that i want to make uh just here sounds

like in brackets

is just that disembodiment does not

necessarily need to be a bad thing right

we usually put things in

good or bad right or wrong but it

doesn't need to be a bad thing because

if we do get some therapeutical effects

um positive effects from psychedelics

this precisely by having this

disembodied experience so it's just a

different way of interacting with the

world and that's why we want to pursue

less of a disorder way and much more of

a there's a spectrum of different

patterns of interaction with the world

right

okay so just that note um

then um i want to look at this

embodiment um

and how we can use active inference to

understand this embodiment in two ways

one is the psycho the psychology of this

embodiment so how that disappointment is

felt psychologically and the other one

is the neurobiology of this embodiment
so as we have said uh this embodiment is
the loss of the body as a subject which
is replaced by
uh some form of
intellectual effort so imagine that you
now don't have these broadly access to
the world that is not possible because
that's exactly what has been lost by
psychopathology or psychedelics intake
and many other things right so then
try to cope and compensate by literally
engaging in making inferences about the
state of affairs so here we could use a
little bit of the literature and
sophisticated active inference which is
this conscious reasoning and planning
and policy selection that you engage now
with having lost these access to the
world through the body
now our hypothesis is that the issue is
that in this disembodied because we all
engage in sophisticated active inference
that's all fine
um the difficulty is that in the case of
this embodiment um experience this
inference is insulated is isolated
because what has been lost is the body
access to the world
so then what happens is um so uh we
would have something that looks like
this so we all know this um this scheme
here right and what we want to say is
that what is lost now is the
influences between sensory and active
states and let me guide you through it
so um let's just uh
focus on the figure for now so we have

internal states as being disembodied reasoning so a subject that has lost this access or bodily access to the world because the body is not felt as a subject so we've got this disembodied reasoning as internal states and then we have external states as the environment where that that the subject wants to explore and interact with and then we have a sensory state the environmental co-constraints so these aspects that should influence uh the subject right and then as active straight states we have the effectivity as um the ways in which the subject can affect and being effect and be affected by the environment but what has been lost here is that there is an unbalance between sensory and active states so for one active states are not sensitive to sensoristics right so it is this effectivity or this possibility of being affected by the environment that is not possible why because the access through the sub the body as a subject has been lost in this psychopathology or psychedelics as that we label as disembodiment experience so um as a coping mechanism what is going to happen is there will be an increase of activity of the activity of active states by increasing the issuing of inferences about the world so i do

not understand

what is going on

but i'm going to continue to think about

it and make and and develop theories

about what is going on

the problem is that we can we all engage

in these developing theories and

inferences about what we think is the

likelihood of the state of affairs the

situation here in this embodiment is

that these inferences are disembodied

so they tend to be false beliefs

and that's because of the insulation of

internal states

what does this mean this means that

there is not um so the access that we

had

with the world through our lived body

is what is lost so then the subject in a

situation where um the only hope is to

increase in increase active states

to make more inferences about the world

to cope with that loss

and then tendency is to have false

beliefs that's when we start entering

the realm of psychopathology and

psychosis

so in these disembodied cases uh this is

actually a very important point because

one could say okay so then the problem

is the is on the level of reasoning so

on the level of for example of the

internal states as we have set up here

no the problem is not on the level of

the internal states reasoning is fine

the inference mechanisms are working

perfectly fine the subject can engage in

in reasoning

it can make
inferences the problem is inside the
markov blanket the problem is in the
unbalance that there is between active
and sensory states because those are the
ones that are supposed to
directly influence one another such that
indirectly there's a connection between
internal and external states but it is
this um direct influence between sensory
and active states
that is problematic they are not
communicating well they are not
influencing uh themselves or each other
in a balanced way
um in a sense and balanced in the sense
that active states become super active
and sensory states um cannot influence
active states so then what we get we get
to this idea of a psychotic markov
blanket which is if we have this sort of
arrangement where there's a problem
between the balance of the strength of
influences between sensory and active
states then we have
a problem of the generation of false
beliefs
the problem being that these beliefs
that are issued by active states they
cannot be updated by the world because
it is precisely this access to the world
that has been lost lost by virtue of
having lost
the body as a subject
so then um
because of this uh this achievement that
we can actually
explain as what is appearing inside the

marco blanket

the reasoning subject makes inference
about inferences about world but um
these reasoning occurs from within the
lack of the experience of the body as a
subject so it occurs in an insulated
manner and this is consistent actually
um with a long-standing literature and
phenomenology of psychosis
um where uh the subjects and sisters a
bunch of children on psychosis linked to
solipsistic behavior
um solid existing uh drawing and
painting and also the philosophical idea
which is the philosophical idea that
only one's mind is sure to exist
and then some some reports that i find
quite interesting is uh uh some patients
who report i was no longer sure
that i was still the same person
i became one with other creatures or
objects i lost a sense of my own
boundaries
when i had an experience i often did not
know it was mine or the experience of
someone else so it is quite interesting
what can be
lost
when we lose these
body as a subject as it is developed in
phenomenology this also happens a little
bit in the personalization disorder um
for example showing this disembodiment
is achievement
when
making people feel estranged and
detached from their self body and the
world

okay so this is all i had on the disembodied on the psychology of this embodiment and now i'd like to turn to the neurobiology and also in the end apply the same framework now to the neurobiology of um of the underlying neurobiology of this environment so we think that applying the same reasoning is consistent so this this achievement that you were talking about felt at agent level should also be expected at the neurological state so that's our hypothesis right so now we think that there are two places there are two good candidates for perception and interception would be good candidates um super perception as a system underlying movement and location of the body in the environment as well as interception as the system responsible for homeostasis these are good candidates we think um for the neurobiology biology of this general disorder that is felt in this embodiment um so proprioceptive and interceptive systems um just like any situated organism can be conceived as self-organizing systems so we apply exactly the same principle as we would apply to the agent as a self-organizing system so the question then that we had is what is the factor that is at the basis of

this of a neurobiological pattern
underlying proprioception and
interception that will show this small
adjustment to the local environment so
if we expect the same sort of mechanism
or organization
dynamics as to the whole
agent and that we were just talking
about then how are we going to find this
out as well in the neurobiology
so we think that
just like in psychological experience of
this embodiment where the agent becomes
insulated from the world and in this
case we say by virtue of losing the body
as a subject the answer will reside in
biological insulation
um now i'm going to focus now on
interception because i don't have time
to also cover proprioception um and i
think that this does give us here some
very very cool insights so um
i'm going to uh focus now on the work uh
brought by baritone siemens on
interceptive predictions in the brain
and they have a vertical model which
they call embodied predictive
interception coding model
and well the goal there is to integrate
an anatomical model of corticocortical
connections with beijing active
inference principles and they propose a
very cool thing which is that a granular
visceral motor cortexes contribute to
interception by issuing interception
predictions
so more specifically what they want to
say is that the granular limbic regions

that regulate visceral motor of the body's internal milieu that is the visceral motor cortices so this is their focus and where do you think they will find the interoceptive system and where they apply the model

now um so you can see the agranular cortex here uh on the left and uh one very interesting thing that i'm just gonna draw your attention to and then if you feel uh that you'd like to read this super interesting paper then um please go for it but i just want to draw your attention to the fact that on the left side you have the agranular cortex and there's one thing that is missing from the agranular cortex as opposed to the these granular and the granular cortex so what is missing is the layer 4 and what happens with that is that there won't be you cannot see there any prediction error neurons so that's pretty cool because that will uh give us an idea that there is no prediction error neurons in the agranular cortex um so just i'm just going to leave that um out there and um just moving uh forward to just like a understanding like the very essence of the embodied predictive interoceptive coding model um they claim that there is an interoceptive system in the brain in which agranular cortices send visceral motor predictions to the body and transmits interoceptive predictions about

the visceral sensory consequences of those predictions so they also tell us that adrenal are visceral motor cortexes estimate the balance between the autonomic metabolic and immunological resources that are available to the body and the predicted requirements of the body based on past experience right so this is going to be very important for us because a chronic imbalance on those things that i was just mentioning can be which can be caused by constantly predicting the need for more metabolic energy to meet the demands for stressors can produce the well-known depression related disruption and it can also eventually down regulate uh provoke a down regulation of hypothalamus or hpa axis negative feedback loops resulting in chronic hyper cortisolemia and in turn can also promote pro-inflammatory state so this is like some of like the problems of having an imbalance on the level of the autonomic metabolic and immunological resources which is what the interoceptive system should be able to do is to give us that nice balance right so then applying exactly the same um the same framework the same model that we applied to the whole body this embodiment or the generalized

feel of this embodiment that is vastly described in the phenomenology literature applying this to the neurobiology of the interperceptive system then what we get is something that can be seen as like quite similar an unbalance between sensory and active state so let me just guide through the figures so here in this figure internal states are interests are or correspond to the interoceptive system and the external stage the environment whichever um the multi-scaled system that receptive system is part of and then we have active stages that a granular quartus that send visceral motor predictions and then we have sensory stage stages the requirements of the body given the history that the body knows to be required for homeostasis so then in this sense what we would have is that active states are not sensitive to sensory states so even though active states can become uh as a coping mechanism much more active and send much more predictions about what is going on what's the state of affairs um because i'm not receiving any information from the requirements of the body so the sensory states so what will happen then because of this insulation is that a false belief is going to be generated so this will correspond to a constantly

prediction of the need for more metabolic energy to meet the demands of the stressors so in this sense we have here another case of what we call a psychotic markov blanket why is it psychotic because it delivers if there is this lack or this imbalance between sensory states or the influences between sensory states and active states then what we should expect is a degeneration of a false belief and we can also expect that at a neurobiological level which is also supported by empirical evidence so yeah this is what i wanted to say uh just thinking um thanking these uh collaborators here that have been thinking about and guiding me through as well these ideas thank you ness really interesting stuff so i guess we'll return to mark and then anybody is yep anyone's free to ask a question in the live chat and i'm writing things down and then after this second piece of the presentation we'll have a discussion so thanks again go for it mark yeah that was great thank you so much wow it's so interesting and i haven't heard some of your stuff about disembodiment yet and you know also this is a research program that i'm extremely interested in so uh i'll be picking your brain at the end of the q a and also after after this event um what might what wasn't made obvious in anez's talk

but i think is implied is that the stickiness the stickiness of these belief systems and what's helping them become insulated is a story about how precision is being estimated and how precision is being deployed that's surely one of the mechanisms that is creating an imbalance between active and sensory states

so you know

how precision is estimated is really the agent's ability to assess the uncertainty of its own priors and when that's working right then those estimations should result in appropriate model updating you know you shouldn't get stuck in an insulated in an insulated situation but rather you should know wow that's not working out the right way and so appropriate updating should occur um and we know that like

maladaptive precision estimation is disastrous right and we're seeing it everywhere in the psychopathology literature today you know schizophrenic delusion is being um characterized as too little precision on sensory data which is putting this heavy emphasis on just top-down predictions so that's why you're seeing the world in a way that isn't being anchored by what's really happening and of course asd is a little bit the flip right too much precision on prediction errors creates a lot of a lot of volatility and a little bit difficulty um establishing good high level predictions

um

and as a quick side note you know the
the kind of insulated insulated um
outcomes that create these maladaptive
uh self-organizing styles they can occur
in lots of ways right so for instance
yeah so addictive substances is one good
way right addictive substances impact
directly precision estimation machinery
right in the way that the addictive
substance engages the dopaminergic
system so addictive substances are
setting precision outright in a way so
no wonder they can create these sort of
sub-optimal styles

but societal expectations do the same
sort of thing you know sub optimally
pinning precision on
expectations about success for example
you know um so you can have cultural
pinning of precision that you should
achieve a certain level of success in
your life and you know we have deep
running relationships between
perfectionism and depression for example
and again for exactly the same reason
that anes said overly high precision on
some sort of expectation would lead to a
constant

pervasive background error as you're not
hitting that high level of expectation
and then just like um feldman barrett
suggests in simmons you have an hpa flip
because the system just doesn't have
unlimited resources um and then the one
that i'm most interested in today
actually is social media which is
turning out to be a like fantastic way

of bending our generative model into all sorts of shapes that cause sub-optimal outcomes right and that makes sense it's just giving us wave after wave of um you know misinformation through filtered images and staged images and of course social media platforms some social media platforms function just like addictive substances or like casinos in a way so again they're hijacking the precision system in just the sorts of ways that cause these problems okay

i just want to quickly summarize before i go on to why what i think the natural mechanisms that keep us out of these problems are so well-being as optimal model generation importantly depends on correct and flexible yeah not these stuck insulated outcomes but correct and flexible self-estimations of uncertainty so that's precision assignments um so misassigning precision to prediction errors which can happen in lots of ways especially in our in our modern world um will inevitably lead to sub-optimal styles of self-organization sub-optimal generative models and with that the production of pathological beliefs and pathological behaviors and um

so that's how it can go wrong or at least that's the starting point it's one of the ways that it can go wrong so that's how it goes wrong then what is it that ordinarily keeps us out of these sorts of bad bootstraps these sorts of

insulated uh disharmonies between the
the system and the environment and one
answer that i like that we like as a
team it was given by alex chance and
colleagues er where they said that
optimal free energy minimizing agents
will benefit from striking a good
balance between pragmatic and epistemic
engagements okay so that is they don't
only act in a pragmatic way that is in
ways that they're highly confident will
reduce free energy over time because
they can be wrong that's part of the
take-home message from some of the
pathological self-organization styles
right they say that you should also a
good free energy minimizing system is
going to balance that against being
driven to epistemically explore and
forage for more information about how
the world works but also about how their
own model works and implied there is
that when they find important counter
evidence while they're out exploring
that they're willing to update their
model relative to that counter evidence
so then the question then if we buy that
and i i think i do buy that's one of the
ways that we stay out of these bad boot
straps and these sub-optimal styles of
interacting well how do we find that
right balance i think that's a good
question if we're gonna ask well how do
we live well what is well-being all
about how do we find that good balance
and the central way that we've been
arguing over a number of papers is that
we literally feel our way to that

balance and um that part of course is predicated on everything working right because if you have an affective disorder then it's going to be exactly that mechanism that's going to be mis-tuned and of course we get all the problems um that anna's just pointed out so um in a series of recent papers we've been exploring the suggestion that um good precision assignment is going to rely in part on the rate at which error is being reduced so that's commonly referred to as aerodynamics and um we're increasingly not alone in this conjecture um we have people like jeff lee and corcelli who gave the first paper on this and of course vanda cruz and then it was beautifully formalized by casper hesp and colleagues um so if you want to look really at the nitty gritty of this that's definitely the paper to check out and if you want to see a great paper on the computational bits of this you should see casper's talk it's there on the slide it's fantastic so um the idea um uh the idea has developed that these aerodynamics yeah the changes in the rate at which error is reduced are registered by the organism as affective states and um if we think of an agent's performance in reducing error in terms of a slope yeah that plots the various speeds that prediction error is being accommodated relative to our expectations then we can think that positively and negatively balanced affective states are a reflection of

better than or worse than slopes of error reduction so um that is just to say we feel good when we're doing better than expected at reducing error relative to expectations yeah better than expected and we feel bad when we do worse than expected at reducing um prediction error and these embodied aerodynamics are important drivers of precision estimation it turns out and i think that makes sense of course right because unexpected increases or decreases in volatility relative to your model is important information for how you should set precision on your beliefs about action policies right so unexpected decreases in the rate at which error is being reduced tells you that your action policy isn't working the way that you thought it was going to work which means you should obviously reduce your confidence in that action policy and the other way as well right if you're unexpectedly doing better at reducing prediction error relative to some policy then you should up your confidence in that policy so the point being here is that precision then isn't adjusted only based on the amount of error or the amount of error reduction but precision is also being tuned relative to the rates at which error is being minimized over time and um that feature of the active inference framework gives us a rather sexy account of momentary subjective

happiness which i think is so
interesting right um momentary subject
of happiness is the result of
unexpectedly reducing prediction error
and that feels good because we are
suddenly doing better than we thought we
would at getting a good predictive grip
on our environment which under normal
circumstances would usually mean that we
are staying healthy and we're staying
well and um juffley and corchelli's
graph makes that clear you know as free
energy is going up um you feel fear and
then you feel unhappy when it's super
high and then you feel hope when it
starts getting better and you feel happy
when uh error is being managed in an
extremely good way but actually it's not
only showing up in active inference
these ideas are being presented in um
in other camps as well rob rutledge is
of course the paradigmatic case um he
and his team have done a number of uh
brain imaging studies that have already
started to show the strong relationship
between subjective feelings of happiness
and doing better than expected in some
particular nation you can see on the
board behind rob there that's the
calculation for happiness which i think
is a bit of a cheeky way to go but i
think it's cool um and uh if you know
the maths then of course you're gonna
you're gonna know what's going on there
too
okay
now we get to the fun stuff this is the
last thing this is sort of we're coming

into the last stretch here's the fun stuff so

we think that the same machinery that underlies momentary subjective happiness is also going to play a leading role in helping us tend towards optimality and to avoid these sort of sub-optimal styles of self-organization so we think it's going to play a special role in an active inference account of what it means to be well um to see

why aerodynamics play that role we just have to ask ourselves well where are the best slopes to be found and the answer is the best slopes are found at the edge of our own predictive capabilities right between what's known and highly reliable and what's unknown and potentially more optimal and when it's all working right

aerodynamics naturally move us out to that edge right where errors are neither so complex that we can't learn anything from them nor too easily predictable that there's no more epistemic value and really the best slope is going to be right there at the edge of your own capabilities and you know that's not a provocative that's not a provocative idea we see that everywhere in the curiosity literature um we know that babies are attracted to medium amounts of complexity not too well known not too volatile we see it in mice put them in a maze with varying rooms of uh grades of complexity and the mouse will tend to hang out in the room that's just above its comfortability and um and you know

we're already building um adaptive and effective robots using these ideas so this isn't this isn't a far out idea so important for our current discussion is that

in order to make the most of those good slopes that are right at the edge of our own capability a predictive system must be sometimes willing to disrupt its own

fixed point attractors yeah so it must be willing to go beyond its habitual policies to go beyond its extremely well-known ways of self-organizing beyond its own homeostatic set points even right and that's one of the roles that aerodynamics should play when everything is working right when it's going well and you're on a good slope of error reduction then you should be motivated to continue along that path but when the niche

thesis to give you productive prediction errors

then the negative valence that comes up that signals to the system that it's time to destroy

some of these fixed point attractors in favor of more wandering policies more explorative styles and i mean if you want to see that in technical terms um just check out uh fristen breaks beer and deco that's the paper to go to so agents that use aerodynamics in this way to set precision on action policies they're going to avoid getting stuck in any one fixed attractor basin for too

long right they'll instead tend to exist
in more meta-stable states more often
they're going to be poised they're going
to be optimally poised in some way
between exploiting already existing off
action policies and performing action uh
information seeking epistemic actions
that reduce uncertainty
and that metastable poise
um
actually grants all sorts of fitness
advantages over
strictly ordered or more chaotic systems
at least in our kind of environment and
that's something that's very well known
in the dynamic systems literature so
again this isn't a very provocative
claim here um and it's precisely because
those sorts of systems have an optimal
balance between you know what you might
think of as efficiency and degeneracy
another way of saying that is um they're
set up to respond efficiently um to
particular contexts those contexts that
are highly predictable but they're also
remaining open to exploring and growing
in a wide variety of other possibilities
okay so notice then that um for a free
energy minimizing system in our kind of
environment
flourishing is not about avoiding errors
i know that's starting to be more
popular but i think it's a really good
point to like hit right on the head
it doesn't flourish by avoiding all
errors
it flourishes by selectively sampling
the right errors

and i think that's so cool like if that
wasn't immediately obvious if you're
still not like you're still not really
catching that point then i think it's
it's a good point to catch soon um
what matters for flourishing for a free
energy minimizing system in our kind of
environment is that it gets the right
kinds of errors because the right kind
of errors into it and of course we know
that you know like outside of the model
it only sounds weird i think inside the
model if you're in that camp outside of
the model we all we already know that
you know overly sterilized environments
make children sick
overly simplified cities make people
depressed like we know too little
volatility um causes problems for
complex self-organizing systems like us
again i don't think it's
such a provocative claim although it can
sound provocative from inside the active
inference framework maybe so finding the
right sorts of volatility and the suite
of cognitive emotional mechanisms that
keep us in touch
with that good volatility uh we think
that that's going to play center stage
in any account of well-being if we're
going to look at it through the active
inference lens and we've done a little
bit of work on this already this is our
aeon article where we do this in a sort
of popular way um if you want to check
it out the title of the article is the
value of uncertainty
good okay uh

last little bit and then we're done oh

so

um in and now we're going to come into

the most speculative part of the idea

and so then we'll stop and then we can

we can talk about it and see where it

goes so in the literature on well-being

um researchers tend to congregate in a

couple of big camps um and two of the

most popular camps are thinking about

well-being in terms of feeling good

that's the hedonic camp and people who

think that well-being is about being

good and that's the eudaimonia camp and

actually those go all the way back to

aristotle featured here as the cool guy

in the glasses so we speculate in our

paper

that actually we might be able to map

those two subjective momentary

subjective happiness and overall

flourishing being a good sort of

predictor on two forms of aerodynamics

um

a local form of aerodynamics and a

global form of aerodynamics

and we think that there's an uh good

reason

to start thinking about how we might

distinguish between local and global

aerodynamics um

because for instance think about video

game design okay so a well-crafted video

game

could be highly tailored in such a way

that it provides you with just the right

evolving changes

to keep you continually making learning

progress relative to the games you're going to keep finding those good slopes and we all know you know good game design does do that it keeps creating these good slopes of air reduction so do good casinos right um but um just like with substance addiction right you can get so immersed in those local good slopes that um you end up surfing them at the expense of the other things that matter in your life so you start neglecting your friendships and your schoolwork and your work um and your overall fitness okay so while you're succeeding locally at the game uh you're not flourishing overall because in fact there's going to be rising volatility across all of these other domains that you're temporarily not paying attention to right so sensitivity to global aerodynamics would essentially amount to being driven in ways that reflect how well we're managing errors across the various domains in our life and as long as an agent was using those more global estimations of how well or poorly they're doing over their various concerns uh to adjust their precision then you're going to have an agent who tends to stay in touch and who tends to move towards optimality relative to those multiple cares and concerns so you're not only improving locally but we think that there's good reason to think that we're also sensitive to how we're

improving overall and that's going to
play a special role in well-being in
particular and um
that makes
if that's right if we're right here then
that makes a really natural fit between
this computational work that we've been
interested in and more recent so-called
network accounts of well-being where
well-being is thought of as the result
of tending to a complex ecosystem of
positive states and positive
achievements so then global aerodynamics
would be crucial here we think to
staying in touch with those various
concerns
in the kinds of way that would allow you
as a whole
to tend towards a good overall optimal
shape um now like i said this is the
most speculative part of the talk nobody
has talked yet about thinking about
aerodynamics happening in a nested way
but i just want to say that we do have
good reason to be thinking along that
line it's unfortunate that lars isn't
here with us tonight because it
piggybacks on some of his deep
parametric modeling so i was hoping to
lay it off on him to really get into the
computational nitty-gritty of why we're
not just
pulling this out of a hat for you um but
we so we build up this idea
that he's already been developing um and
the idea is basically um
he uses a deep parametric model and you
can check out his paper on metacognition

um if there's a way that we can um link
this somewhere i can yeah we can okay
i'll send it to you afterwards um but he
ends up showing that you know there's
higher level policies that are governing
the allocation of precision on lower
level tasks and the kind of cool thing
about this deep parametric model is that
it becomes possible to appreciate
precisions over
those higher level policies over
precision allocation to particular tasks
so you get this sort of nested you get
this nested precision estimation where
at the lower level you have precession
precision being adjusted relative to a
particular task at a higher level you
have precision
adjustments that are also effective we
suspect uh being set over the policies
about how you deploy your precisions
over
all the things you care about so that
that better than or worse than the
global better than or worse then is
going to be all about are you doing
better than or worse than a deploying
precision across your various cares and
concerns not relative to some local
concern so then we think that what
breaks down an addiction in particular
is you start paying attention to
aerodynamics locally in the drug seeking
and taking behaviors and you actually
have a separation of context and we see
that in the brain right you see the
striatum pull apart from the
dorsolateral prefrontal cortex so the

more contextualizing parts of the brain
are no longer communicating in the same
sorts of ways they were before because
you have ultra high precision on these
local aerodynamics so they're good again
is a good example where we're going to
get an insulated sort of psychotic
markov blanket emerge because of an
overestimation of precision because the
the drugs of addiction are are
making that happen so lars and myself
are currently working on clarifying that
aspect of the model so look forward to
work soon on that um and we're looking
at how we might apply that model to
thinking about um meditative development
thinking about attending to attention
precision on precision modeling
precision um
and uh thinking about how those sorts of
mechanisms might help us have better
access to and be better guided by these
higher level precision tuning processes
um yeah so look forward to that and uh
maybe that'll be another thing we can
come on and uh talk about here on the
lab and then hopefully lars will be uh
not an emergency situation that time and
he can come on
great um
so that's probably enough for me up
maybe just as a final note before we
turn it over to q a i might just say
um the future directions for this are
plenty if you buy some of the mechanisms
that we've been thinking about right it
would be cool to start now thinking
about what sorts of endeavors encourage

or disrupt that sort of meta-stable
poise

right meditation hallucinogens
provocative art installations um you
know are they are they helping or
hindering

um can we foster these metastable states
you know and and how you so deep brain
stimulation looks like it's doing
something like this you're getting stuck
in a tractor basin an ocd deep brain
stimulation and julian can talk about
that if it's an interesting point
reopens up the field of affordances by
creating more dynamic neural um
ecosystems

what does this mean for how we structure
our environment if this is right if we
flourish by hanging out at the edge of
our own capabilities then what does that
mean for how we structure our
environment how should we build our
cities i mean abby tabber at um at bath
is doing incredible work on this
thinking about how we might structure
our cities in ways that are
good for us and actually they turn out
to be sort of paradoxically not making
the city easier for us but making it
more challenging getting more diversity
more art installations more guerrilla
gardening uh things because actually the
more complexity we have actually the
better we are as individuals and of
course the biggest thing i'm working on
right now here at uh hokkaido university
is what does that mean for how we design
our technology and how we consume our

technology how about you know what role
is misinformation as a digital crowbar
uh prying us apart from the statistical
realities of our environment what role
is that playing if these flexible fluid
dynamics are part of what it is to be
well that's all that's all future stuff
and uh if you're listening to this
tonight and you think it's a cool talk
and there's something of value here then
uh that's the way i suggest starting to
think next and i'd love to hear anybody
you know here on q a but also um my door
is always open if you uh if you sort of
dig the talk and you dig the research
then definitely reach out because we're
we're always up for collaboration
okay
that's probably good for me you're
welcome
awesome
so fun and so many cool ideas in there
um anyone in the chat is welcome to
provide a question and first i'll just
pass to julian to maybe give any remarks
or
thoughts and then we'll see if there's
any questions in the chat or some
questions i've written down
no i think mark did a great job so let's
just jump straight into the q a
thanks mark
awesome um okay so
you mentioned it kind of tantalizingly
at the end about how we would
differentially be using the technology
and
that just made me wonder like what would

inactive inference

social media look like or how would

somebody of course not see any examples

of it out in the wild how would we even

go about imagining that kind of a system

anybody else want to try to answer first

because i've got something to say if

anything comes up

should i just should i just go

yeah

yeah i mean it's a good question it's

something i'm really interested in um

maybe one thing to say is uh

you might think that uh there's a big

market for making tech

that uh reduces our option set

um you know um that oh well this tech is

great for us because it simplifies our

affordance landscape and that's what

people want they want a sort of

simplified affordance landscape and the

danger is is that actually we're going

to be attracted to that because it's

going to temporarily reduce free energy

faster than we thought it would right

google does a great job of

quickly minimizing our uncertainty about

various topics but over time uh you can

have a sort of nasty outcome and i think

we're starting to see that more and more

where you collapse because it's so

rewarding to come to one place like

you know your smartphone ends up being

one place to come again and again and

again and over time your affordance

landscape is collapsing it's

increasingly collapsing to one and only

one end it's a little bit like walmart

you know it you know you want to buy
there because you can get great deals
but the problem is is that when the
whole community buys there and all the
mom and shop complexities go out of
business uh then you're stuck just with
walmart and then they put up their
prices it's a little bit you know it's
sort of like that you think that the
collapsing

uh tech is going to be good for us but
actually over time having less
complexity is harmful for us and think
about social media platforms as well
like you sterilize social interaction
and it's really rewarding at first
because social interaction is hard it's
a complex affair you know it's very
uncertain how to manage it to be with
another human so temporarily online
interactions can be really really
rewarding because they're so much
simplified but if you hang out there for
too long and you lose some of your
ability to

engage with complex human affairs um
then we're starting to see some of the
ramifications of that sort of technology
i mean we're literally
dying of loneliness even though
we're more
superficially connected than we've ever
been and i think that would be an
interesting project is to start thinking
about

what is working when we have complex
ecosystems of social interactions that
we're losing by just using social media

platforms for our social interaction

yeah julian

yeah so i was thinking about uh

intellectual humility as this is

something we've been talking about

before and

how important it is to

be open to prediction errors if you're

going to get the most out of social

media

so you can think of

filter bubbles as a kind of psychotic

markov blanket in a way and that

there is insulation from any evidence

that doesn't

match with your

your predictions or match with your

expectations

um

so in order to flourish

to to live well with social media

perhaps what's necessary is

is this kind of openness to

uh other contradictory points of view

um

uh what the ancients would have thought

of as humility

and we need to foster that to be well

with technologies

so inis what do you think about filter

bubbles as an example of

a psychotic markov blanket do you think

that's a stretch or does it fit with

your

the way that you describe this this

insulation from sensory evidence that

happens that can induce a kind of

solipsism

right

yeah absolutely i think it's uh exactly
on the right track um actually uh the
way that algorithms are constructed to
deliver precisely these bubbles in
social media

it's so interesting because it's
precisely what's going to happen a
psychological markov blanket because
it's just going to be insulated from
that amount of information making
inferences about that
that precisely precise information
without having any other
contact or exchange outside of the
bubble so that

that's very interesting as well and
there are many many other forms of
psychotic markov blankets that you can
find in the societies and communities
and um that kind of thing

um

but i didn't want to address exciting
things about the concept that it's scale
free

so uh now you see it at collective
levels of self-organization not just
but also the level of the cell you show
so that nmda receptor you can also think
of in terms of

a

yeah psychology yeah

yes i didn't have time to present that
here uh because uh yeah but um but
that's that's also formulated under the
disconnection hypothesis

by carl so it's also really cool because
it's at a cellular level so it's quite

interesting if anyone is interested so
that would be also
what we are um applying or looking at in
the in this paper is another example of
a psychotic markov blanket where there's
a an imbalance between um
again what i was saying the sensory in
the active states formulated there um as
the receptor so and specifically for
schizophrenia but then you can scale all
of these up and find all of these
psychotic markov blankets um in many
different levels and i think that in the
case of social media the bubbles are an
excellent example
but i think that we also have that as
mark was mentioning and i always i
always find it's very interesting uh
when we discuss this topic about um
about how um
specifically um the
city living
as an environmental
daily living space
influence
mental health and
and
and specifically
why and when some groups of individuals
thrive in an urban setting so this is
something that some people here in
amsterdam for example are working on and
it's really interesting um and i think
that it becomes even more interesting if
we come for example from
studies in neurodiversity for example in
autism and that kind of thing and if we
think about the design of these spaces

um daily spaces daily living spaces that
um are not designed for example for
diverse ways of interacting with worlds
such as an autistic
interacting with the world so i think
that um it is very relevant to think
about
these
mental problems and how we can inform
interventions in the city planning and
also
smart environments um
that would allow a more diverse way of
interacting with the environment as
opposed to reducing the complexity of
the environment which is exactly what
mark was talking about and i just wanted
to add that um to that
yeah we're actually working with a
pretty cool group through the nested
minds network um developing models of
anti-fragility and what we're calling
trophilia which is the love of
uncertainty so you're not just growing
from volatility but you're actually
seeking volatility in order to grow and
we're thinking about it at the level of
communities so um we're still just
getting started but um it's a really
fascinating really fascinating field
yeah so one comment on that and then a
question from the chat um julian you
pointed out that active inference as a
scale free and sort of uh base system
agnostic framework really helps draw out
these similarities across systems and
then mark in the in the presentation you
talked about neural ecosystems about

social social media ecosystems and then thinking about some of these pathologies or perceived pathologies in an ecosystem case i mean imagine if someone said i like this tree this should be the only tree or i'm seeing a little tension here with a predator and the prey we should separate those two species those would lead to a catastrophic collapse because it wouldn't be the frat it wouldn't be the anti-fragile ecosystem that was the one that had made it all the way to the present moment and was existing far from equilibrium so it's sort of like when we can look across systems it gives us a new way to look at the systems that we can modify the nisha of like our cities the internet and then starts hopefully shining some light on where we can look here's the question from the chat dean wrote remark a simple provisioning rule when in doubt zoom in comma zoom out so that's dean's providing that is sort of like a little simple uh motto as an attending strategy could this be a happy person setting up a local or global non-isolating markov blanket like how do we go from these mathematical framings and some of these aerodynamics into nice slogans like dean does no i love that i mean right yeah exactly yeah look look close and look far i mean that's that sounds like that sounds like

super good advice

um a little bit you know this a little
bit follows on from the same point
julian was making earlier about virtue
about some of the like what would the
virtues be

uh

relative to this kind of good updating
system and um i've been meaning to do a
i've been meaning to write something on
this for a long time now and you know
i'm getting more and more motivated as
we go along but have you ever heard of
super forecasters have you heard of this
it's a book out called super forecasters
so there's people around the world who
go to competitions and they predict
things from noisy data sets and you can
train in it and you get in a team you
can do individual events and people can
get better so you can train to be a
super forecaster so you take noisy data
you get good at predicting it and i like
this idea because i think it's a nice
metaphor you know i think we might be
able to glean like if we're all
predictors

and we're trying to predict well then
maybe there's something to learn from
people who try to predict well as a
hobby you know and try to see like what
are they doing right um and there's some
really interesting things that they do
uh that i think are um
useful given the kind of framework that
we're proposing here a couple of them i
think just everybody should know about
one a super forecaster across the board

the best the best super forecasters in
the world they share a couple of common
traits they are curious
so they're always interested not just in
their own field but they're interested
in lots of field they tend to be
polymaths so they're interested in lots
of different sorts of things that's one
thing the second thing and this is a
little bit i feel like is related to
this zoom in zoom out point and also
exactly what julian said is that they
have they have they hold so this active
inference language but they hold
high-level beliefs
that all of their beliefs are only
relatively right

so

um

they know that when they're looking at
something not to get too enamored by it
because it's up for updating at any time
so they're always checking it against
the other things they know they're the
kind of person that when you meet them
in public and you tell them something
they don't know they don't lock down and
say no no i can't be like that no no you
know they're the kind of person who says
wow wow i didn't know that tell me more
about that so i can update i can update
my overall model
relative to how the world is
anyway i think i think these like
this this approach to zooming in and
zooming out sounds just right and i
think it's so cool to start thinking
about what some what are some of the

practical virtues that would help us not
get stuck and i think uh yes
follow up on that so yeah i think what
you just described to the super
forecasters is
something that goes back to the ancients
again the peronean skeptics who put
forward the idea that we shouldn't
do we shouldn't endorse a
proposition but nor should we withhold
our endorsement we should rather try to
find this balance
and what you're describing with the
super forecasters with that kind of
hybrid project i think is exactly that
same kind of
uh
balance well that's the paper peyronian
skepticism
through the lens of active inference
that's
realm it made me think about skepticism
and like almost like a bayesian stoicism
like okay the info has come in through
the channel it came in from i would like
to learn more i will marginally update
like that sort of approach and then also
another
dialectic that you brought up that it'd
be interesting to hear a little more
about is that hedonic short-term
happiness
versus that um the eudemonia the longer
term happiness just like that's super
fascinating to see how
you mapped
two pillars of philosophy really going
so far back

and

integrate them in in a new way like

where do we go from there how do we

think about these kinds of cool

synthesis between philosophy and

computational dynamics

i love that julie do you want to say

anything on that

if not i could maybe start and you could

follow on what do you think yeah so uh

the way we were thinking about it was

though to take i don't know if mark did

this example in the talks i had to pop

out but the computer game example did

you go through that mark yes exactly you

know you might you might find a

a computer game that gives you just the

right amount of challenges to keep you

constantly

making progress in

playing the game

but still in the rest of the person's

life

going to school

uh having your friendships grow

your family life all of these things

could be suffering

so

they're just finding local pleasure

which we can assume the computer game

might continually give the person

by presenting them with this constant

challenge

in the end wouldn't be conducive to

flourishing for that person because all

of the other aspects of their life that

are also valuable and important to them

would end up being neglected

so thinking about that from a a
dynamical or a computational perspective
it seems what's important
is a kind of psychological flexibility
or
and that
you're able to
be poised between
exploiting and exploring
so the person who's playing the computer
game is really just stuck in a kind of
exploitative mode
where
that's continually
um
yielding something rewarding for them
because they're continually presented
with challenges that they can solve
but at the same time they're not
exploring they're not curious
and so there's not that kind of
psychological flexibility which we think
is required for
eudaimonia or
really living a good life and
flourishing as a person
and so we think what you get out of
active inference then is is a
neuro computational frameworks a
biological story but also a
computational story
about uh what it might be to to as a
person live a good life
what might be required is this kind of
openness and flexibility
yeah yeah it's great
two other quick things on the tale of
that one is

you know there's a very old debate about higher and lower hedonia
yeah the low hedonia is the sort of animal life that causes all the problems higher hedonia but there is higher hedonia that leads you to whatever the good life is that's kind of interesting from this perspective as well because you can think about it i think maybe as lower hedonia is aerodynamics in niches that lead you away from the sort of metastable attunement
right so drugs of addiction is a good example of lower hedonia because it's good prediction error slopes but they're only local at the expense of your more integrated holistic optimality of governing
state right but what would a higher hedonia be well it would be a local good slope that leads you to a greater integration
so at the tail end of our paper and i didn't mention it here because it's a bit of a further topic but we suggested maybe thinking about zero sum and non-zero sum activities and their role in producing or restricting these sorts of metastable attunements and i think that's a good example where you can get you can get hot about some small thing that reduces your dynamicism so you get this global rising in air even though you're locally succeeding that's a nice i think it's a nice way to think about lower hedonia higher hedonias local good slopes that

encourage and expand and and generate a
richer and more complex uh ecosystem of
good slopes in your life so for instance
like
well like think about the things that
are endlessly rewarding
you know love
relationships you know like you're never
going to get to the end of being a good
husband or wife you're never going to
get to the end of being a good older
brother or being a good dad good mom
you're never going to get to the end of
those that's that's an endless thing or
patience patience compassion love
you know sympathy empathy i mean you're
never going to get to the end of those
things service service to the people
into your community service is such a
good example here if you feel good like
if you get good slopes in serving your
community you're just going to keep
creating more connections
that are going to create more
opportunities for new kinds of slopes to
be caught and encouraged so i think it's
it's perfect almost you know it's a
local slope that leads to lots more
other stuff and just one other quick
thing and i won't i won't i won't harp
on about it but
uh and i think also that uh a system
that's set up to be
uh to have this more global aerodynamic
regime working correctly i think they'll
also tend to have more subjective
happiness
and it would be good to like go through

this and build a
detailed computational model because
this part is just a little bit
speculative but it seems to me that if
you have a really rich diverse ecosystem
of good slopes then if you reach the end
of some particular slope like let's say
you lose the video game if the video
game is the only thing you have
no wonder we have a problem regulating
our emotions if you know you see this on
youtube all the time you know mom or dad
turns off the system and there's an
explosion and they say so this is the
only thing i have right so you're very
very rigid you're very fragile with only
one outcome but if you have lots of
stuff going on in your life then if you
lose a good slope here it's just a
matter of task switching over to the
next good slope and so i tend to think
that people who have a diversity of good
slopes in their life of course they're
also going to have
more persistent more pervasive good
feelings
but you know that's a bit speculative so
love the look
this local global nested slopes descent
i think it really helps us understand
active inference but also understands or
helps us understand how to apply active
inference because it doesn't have to be
an active inference model about emotion
but that's the one we're talking about
but it could be some other kind of
active inference model so kind of just
two other um concordances i guess

uh cezanne mahali's flow and flow
psychology it's like you mentioned
selecting the right kind of errors and
so it's like there's there's being in
the flow state of playing chess or of
doing a skill and then there's like
getting the rock out of your shoe it's
like you're you're still in this sort of
uncertainty regime but one of those is
just delaying you on the journey and the
other one kind of is the journey so that
and then one other
it's dead on it's dead on actually john
verveigh at the university of toronto
and i are planning a paper
using this exact model to think about
flow as cascading cascading insight so
every time you reduce you hit that good
slope and then you learn that's what the
good slope is you're learning and when
you get to the end of that learning what
it does is it opens up a new good slope
that you learn down which opens up a new
good slope which you learn down and that
happens in a sort of cascading a
cascading pattern yeah i love that
so that that
perfectly comes to the second example
the cascading insight like so many
people whether they're in academia or
just curious about something
when they get a quote answer they say
well you know just brought up so many
more questions like now i know how much
this planet is in diameter but now i'm
curious about something else and so it's
like that that short term minimization
is like disciplinary research like we

had 25 efficiency on this chemistry
extraction we got the 27 it's a local
who could who can say that's not a local
improvement but then that global
is the transdisciplinary understanding
and as the research ecosystem if we just
dove down local disciplinary rabbit
holes we would end up with just a wreck
but at the same time we do need to do
some local gradient descent otherwise
we're not extracting that compound so
what can we build our transdisciplinary
theory on and so it's just like
very cool how we can um have a positive
um i don't know can it be neutral and
positive at the same time
yeah yeah why not yeah it's positive in
the sense that it's not negative uh i
but i i love that and i think that's a i
think it's a role that philosophers are
increasingly um playing you know
what you might call synthetic
philosophers um philosophers who are
trained in various scientific
disciplines the neighboring scientific
disciplines around a certain topic like
dan dennett you know who are able to
zoom out and create good overall
frameworks that sort of uh that bring
together various disciplines as well as
the special sciences that zoom in that
zoom in and uh discover locally they
come in on this because i think one of
the interesting thing is the three of us
here are all philosophers so
we will come at active inference through
philosophy
but through trying to do that synthetic

kind of philosophy where
we're integrating philosophy with
neuroscience with biology
and
so that's a common theme and another
thing that i think is interesting among
the three of us is that we're all
thinking about active inference as
something that
is environment involving so one of the
crucial interesting things i think in
innis's work on
the psychotic mark off blankets is what
happens when you are
you're you have a
a free energy minimizing system that's
insulated or isolated from
what's coming in from the environment
and how that can induce pathologies
and a lot of the things we've been
talking about today are about kind of
designing an ecosystem
where
being in that kind of environment would
help you to flourish as an individual so
what would uh how would you need social
media to look
if it was instead of creating
the kind of social pathologies that
we're seeing today
with the misinformation problems and
all of the problems that are arising
from polarization you had a a social
media environment in which people
flourish
and so we think that
these ideas about active inference
if you have a philosophical perspective

on them you can see how they
they could
help you to
know to think about ethical questions
maybe political questions as well
it's about synthetic perspective that
philosophy can bring
helps you to see how to
to take this framework and
and
show how it can be applied in lots of
different domains
i think that was something that you
touched on daniel in one of your
observations questions
awesome man i think it's a valuable
contribution that philosophers can make
from from using these ideas
totally we see advances in philosophy
alongside and deeply interwoven with
governance and technology so here's just
one
last question i wanted to ask on that
theme unless anyone has any other
questions and this is um to and as his
section so you talked about
thinking about using active inference as
a model which was invoking this realism
instrumentalism
distinction that we had talked about in
some other streams
and i just wondered how will realism
versus instrumentalism or potentially
other philosophical ideas play out
clinically like
what will it look like to
deploy something instrumentally and say
that for example this is being supported

by a mathematical framework but that's not what is per se happening inside of you it's just how we're instrumentally interacting with it i don't know just how will that kind of a philosophical nuance

enter into the regular protocols of medicine or governance

um yeah um so i think that um

instrumentalism is very much aligned with dynamical systems theory take on cognition or on brain activity

right so it doesn't need to be something that um

is highly mystical or that is like a framework that we use but then what can

we do with it really like in in medicine or in more experimental settings right

it's just that it's not um committed to a more computationalist

account

of cognition or of the brain it's much

more aligned with for example in this

framework with the dynamical systems theory as in the case of being

a tool that is useful to

explain and describe um

an activity of scientific interests

through some constructs that we know are

constructs that are we build for our own

scientific practice and understanding

right so that's the idea

so um

it is useful in the sense that you can

you can now put things together and make

this activity that would otherwise be a

blur intelligible in a scientific sense

by using these constructs which you can

think about it for example from
dynamical systems theory
mark was mentioning these attractors for
example which by the way um i'd like to
say something about that um
um in the case of um for example in
psychopathology there is this kind of
like fixed points or attractors that
kind of like we need to change etc we
are not saying that there is actually a
fixed point or an attractor
in the psychopathology but we are
explaining
this in this sense for our own or to
make these patterns intelligible
from a scientific point of view
right so it's in this sense that you can
use this instrumentalist account to say
that
you gain explanatory attraction but
you're not saying that actually there is
a fixed point in the behavior itself
does this help
yeah i think so i mean if someone looks
at an economic model and it uses a
regression it's not saying the economy
is a regression but then there's
something about
the the personal nature of the brain and
our emotions
where it feels almost like a model of
emotion or emotional dynamics
it's something that's been brought up
many times i mean a lot of people like
qualitative perspectives on these topics
because it
it averts this whole discussion on
whether the mathematical models are real

or instrumental and it almost feels like
now with that philosophical clarity
to be able to frame the two different
modes we have there we can distinguish
very clearly

instrumentalism from a sort of
background of
implicit realism and then that would
maybe help speak to a broader set of
people who have different beliefs and
different structures for what they think
is the underlying system

can i comment on this briefly yeah
please

yes i i think i take a different
perspective reminisce on the realism
instrumentalism issue

so i i start from the cybernetic idea of
the good regulator theorem that ashby
put forward and

i think that gives us a way of thinking
about how values are actually built into
living systems

and so the idea of being a model of the
environment it's not just having a model
but actually being a model

and how you can be a good model
so what does it take to be a good model
while there has to be as much complexity
in the model as there is in the
environment that you're interacting with

so i think what we get out of there
um

out of ashby's good regulated theorem is
an idea about what it is to be
well adapted to the environment
and i don't take that just to be like a
model of

uh of life or of systems that are well adapted but that we can actually get some principles that tell us how would a system need to be organized in order to be well adapted

um and those principles yeah i mean they are a model i agree

but they actually tell us something about the organization that a complex adaptive system would need to have if it was to remain well adapted over time so if it was to be a good model and then we can get some kind of value or normativity out of that because notice that there's a notion of goodness implied here that we're thinking about what would it take for a model to be a good model of its environment for it to remain well adapted when there's a kind of value or normativity implied there as well

so i think all of that is not just there about you know finding explanatory models that do a good job for us as scientists but it's also telling us something about what it is to be alive what it is to be a good cognizer so that we could begin to get some kind of values out of biology and i take that to be a kind of realist project

and
because
you know i think that
we can
we can get some motion or value
ourselves
free energy minimization in fact i think
the notion of free energy is itself a
kind of value for a system that tells a
a system what it is to and to be well
adapted it's well adapted if it
if it is first of all minimizing free
energy over time but we saw from marx
talking from our paper that on its own
isn't sufficient
you need in addition this kind of
maintaining less stable poise
uh so now all of that is just to say
well i think we can get some
um
some something more than just a model
out of free energy free energy principle
we can get
some principles which help us to make
sense of how there could be values in a
in a world of facts
and so it helps us with that question it
may help us to understand
you know
what it is to live a good life so
and how could you have
natural systems that flourish
so we could get some kind of
um naturalistic account of
uh of ethical values
now so that would be my take on it and i
think that's kind of different from from
instrumentalism a different kind of

perspective on instrumentalism i think

you can also get out for free energy

principles

there's a debate here isn't there and

that's one of the things that philosophy

can bring in there's different

perspectives you can have on

what the free energy principle means

and then we can fight about that amongst

ourselves as philosophers

great closing point we respect and value

the uh

the content but then also there's that

hyper philosophical prior that the space

for the debate and the space for

different perspectives is itself

important so with that

thanks to all of you this was an awesome

session you're always as well as anyone

else welcome to come back share any

other exciting research or questions

so

great times and i hope everyone watching

live and replay appreciate this great

session so thanks again everyone and

see you next time thank you for having

us thanks so much daniel thanks everyone

thanks so much